No-Arbitrage Does Not Imply an Equivalent Calibrated Pricing Measure

Equilibrium Daily

Abstract

Nutz, Wiesel and Zhao (2023) assume that some pricing measure is equivalent to the physical law, reprices every European option and makes the stock a martingale, and they leave its characterization in terms of trading as an open problem. Ordinary no-arbitrage does not characterize it. We construct a two-date market with bounded prices, a physical law that is itself a martingale and option-implied terminal probabilities within a factor of two of the physical ones, in which there is no arbitrage but every calibrated martingale measure assigns zero probability to an intermediate state of physical probability one half. In a family of such markets, no-arbitrage requires a weighted series to diverge, while equivalent calibration requires a uniform lower bound on the same weights. Adding a closure condition restores the characterization, through standard separation and entropy-duality arguments.

1 Question

Let P be a probability on $\mathbb{R}^2$ with coordinates (X, Y), the stock price at an intermediate and a terminal date, and let ν be a probability on $\mathbb{R}$ with $\int |y|\,\nu(dy) < \infty$, the terminal law implied by option prices. A strategy is a pair of finite-valued Borel functions h and $g \in L^1(\nu)$, with zero-cost gain

$$V(h, g) = h(X)(Y - X) + g(Y) - \nu(g), \qquad \nu(g) = \int g\, d\nu. \tag{1}$$

The option position g is bought at its implied price $\nu(g)$, and h is the stock holding chosen at the intermediate date. A gain is admissible if $V \geq -a$ P-a.s. for some constant a; no other bound or integrability condition is imposed. No-arbitrage (NA) means that every admissible $V \geq 0$ is zero P-a.s. Write

$$\mathcal{M}(\nu) = \{Q \sim P :\ Q_Y = \nu,\ \mathbb{E}_Q|X| + \mathbb{E}_Q|Y| < \infty,\ \mathbb{E}_Q[Y \mid X] = X\}$$

for the equivalent calibrated martingale measures, and $\mathcal{M}_{\text{fin}}(\nu)$ for those with finite relative entropy $H(Q \mid P)$. Nutz, Wiesel and Zhao (2023) take $\mathcal{M}_{\text{fin}}(\nu) \neq \varnothing$ as a standing assumption and describe its characterization in terms of trading strategies, or even that of $\mathcal{M}(\nu) \neq \varnothing$, as an open problem. If $\mathcal{M}(\nu) \neq \varnothing$, every admissible gain has zero Q-mean, so NA holds. The question is whether the converse holds.

2 No-arbitrage is not enough

Put $\ell = \log 2$ and, for $n \geq 1$,

$$y_n = 2 + (-1)^n(1 + 2^{-n}), \qquad x_n = \tfrac{2}{3}y_n + \tfrac{1}{3}y_{n+1}, \qquad x_0 = \tfrac{1}{2}(y_1 + y_2) = \tfrac{15}{8}.$$

Row $n \geq 1$ is the two-point law $R_n = \frac{2}{3}\delta_{(x_n, y_n)} + \frac{1}{3}\delta_{(x_n, y_{n+1})}$, and row 0 is $R_0 = \frac{1}{2}\delta_{(x_0, y_1)} + \frac{1}{2}\delta_{(x_0, y_2)}$. Each row is a martingale law. With weights $t_n = 2^{-n}/(n\ell)$, which sum to one,

$$\bar{Q} = \sum_{n \geq 1} t_n R_n, \qquad \nu = \bar{Q}_Y, \qquad P = \tfrac{1}{2}\bar{Q} + \tfrac{1}{2}R_0.$$

Theorem 1. *In this market NA holds for all finite-valued h, including unbounded ones. Nevertheless $\mathcal{M}(\nu) = \varnothing$. The only calibrated martingale measure absolutely continuous with respect to P is $\bar{Q}$, which gives zero probability to the event $\{X = x_0\}$ of physical probability $\frac{1}{2}$. All prices lie in $[\frac{1}{2}, \frac{13}{4}]$, P is itself a martingale law, the rows form a connected chain, and $\frac{1}{2} \le dP_Y/d\nu < \frac{3}{2}$.*

Proof. No equivalent calibrated measure. A martingale measure $Q \ll P$ must put the row's own branch probabilities on each row, so $Q = s_0 R_0 + \sum_n s_n R_n$. Write $\delta_n = s_n - t_n$. Calibration at y_1, at y_2 and at y_n for $n \ge 3$ gives

$$\tfrac{2}{3}\delta_1 + \tfrac{1}{2}s_0 = 0, \qquad \tfrac{2}{3}\delta_2 + \tfrac{1}{3}\delta_1 + \tfrac{1}{2}s_0 = 0, \qquad \tfrac{2}{3}\delta_n + \tfrac{1}{3}\delta_{n-1} = 0,$$

so $s_1 = t_1 - \frac{3}{4}s_0$ and $s_n = t_n + \frac{3}{2}(-1)^{n-1}2^{-n}s_0$ for $n \ge 2$. For even n this is $s_n = 2^{-n}(1/(n\ell) - \frac{3}{2}s_0)$, which is negative once $n > 1/(\frac{3}{2}s_0\ell)$ unless $s_0 = 0$. Hence $s_0 = 0$ and $Q = \bar{Q}$.

No arbitrage. Let $V = V(h, g) \ge 0$ P-a.s., and write $c = \nu(g)$ and $g_n = g(y_n)$. Within row n the stock term has zero mean, so $b_n := R_n(V) = \frac{2}{3}g_n + \frac{1}{3}g_{n+1} - c \ge 0$. Since $g \in L^1(\nu)$, $\sum_n t_n|b_n| < \infty$ and $\sum_n t_n b_n = \nu(g) - c = 0$, so every $b_n = 0$. This gives $g_{n+1} - c = -2(g_n - c)$, hence $g_n - c = (-2)^{n-1}(g_1 - c)$. Because $\nu(\{y_n\}) \ge \frac{2}{3}t_n$, $\sum_n \nu(\{y_n\})2^{n-1} \ge \frac{1}{3\ell}\sum_n 1/n = \infty$, so integrability of g forces $g_1 = c$ and then $g_n = c$ for all n. On row 0 the gain reduces to $h(x_0)(y - x_0)$, which takes opposite signs at y_1 and y_2 unless $h(x_0) = 0$. Thus $V = 0$ P-a.s.

Regularity. Direct computation gives $\frac{1}{2} \le y_n \le \frac{13}{4}$, distinct x_n different from x_0, $\nu(\{y_1\}) = 1/(3\ell)$, $\nu(\{y_2\}) = 1/(4\ell)$, and $P_Y = \frac{1}{2}\nu + \frac{1}{4}\delta_{y_1} + \frac{1}{4}\delta_{y_2}$, so $dP_Y/d\nu$ equals $\frac{1}{2} + \frac{3}{4}\ell$ at y_1, $\frac{1}{2} + \ell$ at y_2 and $\frac{1}{2}$ elsewhere. □

The obstruction comes from matching the whole terminal law. On each row the physical conditional law is already the required martingale law, but calibration to ν propagates a perturbation of row 0 along the chain with alternating sign and geometrically shrinking size, while the even-row weights shrink slightly faster. An arbitrage that exploited this would need an option payoff that doubles at each step down the chain, which is not ν-integrable.

Acciaio, Larsson and Schachermayer (2017) show that the set of semistatic outcomes need not be closed, and Nutz, Wiesel and Zhao (2023) point to this failure of closure when they leave the question open. Theorem 1 shows that the same mechanism breaks equivalent calibration for a fixed physical law, and that it does so in a regular setting, with bounded prices, a physical law that is itself a martingale, and physical and option-implied terminal laws within a factor of two of each other.

The same chain with general weights shows how far apart the two conditions are. Keep the rows and let (t_n) be any strictly positive probability sequence.

Proposition 1. *In the chain with weights (t_n),*

$$NA \iff \sum_{n\ge1} 2^n t_n = \infty, \qquad \mathcal{M}(\nu) \ne \varnothing \iff \inf_{k\ge1} 2^{2k} t_{2k} > 0.$$

Proof. If $\sum 2^n t_n = \infty$, the proof of Theorem 1 applies unchanged. If the series converges, set $g(y_n) = -(-2)^{n-1}$, which is ν-integrable with $\nu(g) = 0$. Every b_n is zero, and on row n the single holding $h(x_n) = -g(y_n)/(y_n - x_n)$ sets the gain to zero at both atoms, because the two atom gains have the same ratio as the two branch distances. On row 0, $g(y_1) = -1$ and $g(y_2) = 2$; since $y_2 - x_0 = -(y_1 - x_0)$, choosing $h(x_0)(y_1 - x_0) = \frac{3}{2}$ makes both gains equal to $\frac{1}{2}$. This is an arbitrage. For calibration, the recursion above holds with $s_0 = s$, and an equivalent law exists if and only if some $s > 0$ keeps every s_n positive. Odd rows impose only $s < \frac{4}{3}t_1$, and even rows impose $s < \frac{2}{3}\,2^n t_n$, which some $s > 0$ satisfies for all even n exactly when $\inf_k 2^{2k} t_{2k} > 0$. □

The weights $t_n = 2^{-n}/(n\ell)$ of Theorem 1 make the series diverge like the harmonic series while $2^{2k}t_{2k} = 1/(2k\ell)$ tends to zero. If P has finitely many atoms, NA does imply $\mathcal{M}(\nu) \ne \varnothing$, with finite entropy, because the attainable gains then form a finite-dimensional space and separation gives a strictly positive state-price vector. The example uses infinitely many terminal states linked in a single chain.

3 What restores the characterization

Let $\mathcal{A}$ be the admissible gains and $\mathcal{C} = \{f \in L^\infty(P) : f \le V \text{ for some } V \in \mathcal{A}\}$ the bounded claims they superhedge. The one market-specific step is to identify the measures that price $\mathcal{C}$ nonpositively.

Lemma 1. *For a probability $Q \ll P$, $\mathbb{E}_Q f \le 0$ for all $f \in \mathcal{C}$ if and only if $Q_Y = \nu$ and $\mathbb{E}_Q[Y \mid X] = X$ with $\mathbb{E}_Q|Y| < \infty$.*

Proof. Bounded static positions $\pm(g(Y) - \nu(g))$ lie in $\mathcal{C}$, which gives $Q_Y = \nu$ and $\mathbb{E}_Q|Y| = \int |y|\, d\nu$. For a Borel h supported on $[-n, n]$ with $\|h\|_\infty = M$, the gains $\pm h(X)(Y - X) + M|Y| - M\nu(|y|)$ are admissible, since they are bounded below by $-M(n + \nu(|y|))$. They are Q-integrable and have nonpositive mean, so $\mathbb{E}_Q[h(X)(Y - X)] = 0$, and letting $n \to \infty$ gives $\mathbb{E}_Q[Y \mid X] = X$. The priced straddle $|Y|$ is what makes localized stock positions admissible without any moment condition on P. Conversely, let Q satisfy the stated conditions and $V = D + g(Y) - \nu(g) \in \mathcal{A}$ with $D = h(X)(Y - X)$. The lower bound gives $D^- \le a + |g(Y)| + |\nu(g)|$, which is Q-integrable. Conditionally on $X = x$, the law of Y has finite mean x and $h(x)$ is a finite number, so the conditional expectations of D^+ and D^- are equal and finite; by Tonelli's theorem $\mathbb{E}_Q D^+ = \mathbb{E}_Q D^- < \infty$. Hence every admissible gain has mean zero and $\mathbb{E}_Q f \le 0$ on $\mathcal{C}$. □

Remark. With Lemma 1, two characterizations follow from known results, and we do not claim them as new. By the Kreps-Yan theorem (Kreps, 1981; Yan, 1980), $\mathcal{M}(\nu) \neq \varnothing$ if and only if the $\sigma(L^\infty, L^1)$-closure of $\mathcal{C}$ meets L^∞_+ only at zero. For finite entropy, let $R(W) = \inf_{V \in \mathcal{A}} \mathbb{E}_P[e^{(W-V)^+}]$ be the smallest expected exponential shortfall of a bounded liability W. Then $\mathcal{M}_{\text{fin}}(\nu) \neq \varnothing$ if and only if $R(\mathbf{1}_A) > 1$ for every event A with $P(A) > 0$, that is, no sequence of admissible trades insures a loss on A so well that its expected exponential shortfall approaches the no-loss value. The proof takes a subgradient of the convex, monotone, norm-continuous function R, uses the variational formula for relative entropy to show that it is countably additive with a finite-entropy density, and applies Lemma 1. This is a version of the known link between finite-entropy martingale measures and exponential utility (Frittelli, 2000; Delbaen et al., 2002).

4 Summary

For semistatic trading with an unrestricted, fully priced option menu, ordinary no-arbitrage is necessary but not sufficient for an equivalent calibrated martingale measure. The counterexample (Theorem 1) and the exact gap between the two conditions in the chain family (Proposition 1) are the contribution of this note. The closure characterizations in the Remark apply standard separation and entropy-duality arguments to this market. Whether a simpler support or kernel condition characterizes equivalence beyond the finite-state and chain cases is open.

References

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